ADMM CASE STUDY

Sales & Revenue Analysis for a Small Business

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I. INTRODUCTION

1.1 Introduction:

TrendMart is a small business that maintains customer and sales transaction data containing information about customers, products, categories, quantities, prices, transaction values, locations, and payment methods.

The objective of this case study is to transform the available sales and customer data into a structured analytical solution that supports **sales analysis, product and category performance analysis, customer purchasing behavior, customer segmentation, revenue analysis, sales forecasting, and predictive customer value analysis**.

The solution follows an **ETL and Business Intelligence architecture**. The source CSV datasets are extracted and explored using **Python in Google Colab**, cleaned and validated, transformed into analytical datasets, and loaded into a **MySQL database**. The warehouse data is then connected to **Power BI** for interactive dashboards and reporting.

Python was also used for **sales forecasting and predictive customer value modeling**. The final Power BI solution provides a consolidated view of sales performance, product performance, customer behavior, customer segments, historical customer value, and predictive insights.

1.2 Business Problem & Objectives:

- Business Problem-

TrendMart has transactional sales and customer information, but the raw datasets need to be organized and analyzed to understand business performance effectively.

Without an integrated analytical solution, it is difficult to efficiently identify:

- Overall sales and revenue performance
- Monthly sales trends
- Best-performing products and categories
- Customer purchasing frequency
- Customer revenue contribution
- High-value customer segments
- Regional and gender-based revenue patterns
- Future sales trends
- Future customer value patterns

The case study therefore focuses on building an analytical pipeline that converts the available customer and sales data into meaningful business insights.

- Objectives-

The key objectives of the project are:

- **Extract** customer and sales data from the available source CSV datasets.
- **Explore** the datasets using Python and Pandas.
- **Validate** data completeness, duplicates, data types, relationships, and numerical consistency.
- **Clean and transform** the source data into analytical datasets.
- **Load** the transformed data into MySQL.
- **Perform SQL-based business analysis** on sales, products, customers, locations, payment methods, and customer segments.
- **Create** an integrated Power BI data model.
- **Develop** DAX measures for key sales and customer KPIs.
- **Analyze** product and category performance.
- **Analyze** customer purchasing behavior and segmentation.
- **Calculate** historical transaction-based customer value.
- **Develop** a sales forecasting model using Python.
- **Develop** a short-horizon predictive customer value model.
- **Create** interactive Power BI dashboards for business analysis.

II. Architecture Overview

2.1 Architecture Overview

The TrendMart analytics solution follows a structured **ETL, Data Warehouse, Business Intelligence, and Predictive Analytics architecture**. The process begins with the source customer and sales datasets and progresses through data exploration, cleaning, transformation, database loading, SQL analysis, Power BI reporting, and predictive analytics.

The major components of the architecture are:

1. Source Data

The project uses two CSV datasets:

- customers_updated.csv
- sales_updated.csv

The customer dataset contains customer master information, while the sales dataset contains transaction-level sales information.

2. Python / Google Colab

Python and Pandas were used for:

- Data loading
- Data exploration
- Data profiling
- Missing-value analysis
- Duplicate analysis
- Data validation
- Data transformation
- Customer-level metric generation
- Sales forecasting
- Predictive customer value modeling

3. ETL & Cleaned Data

The source datasets were cleaned and transformed into analytical CSV files.

The main cleaned datasets generated were:

- `TrendMart_Customers_Cleaned.csv`
- `TrendMart_Sales_Cleaned.csv`
- `TrendMart_Customer_Metrics.csv`
- `TrendMart_Sales_Forecast.csv`
- `TrendMart_Predictive_CLV.csv`

4. MySQL Database

The transformed datasets were loaded into the **TrendMart_Analytics** database.The main analytical tables created were:

- `Customers`
- `Sales`
- `Customer_Metrics`

5. Power BI

Power BI was used for:

- Data modeling
- Table relationships
- DAX measures
- KPI development
- Interactive visualizations
- Sales analysis
- Product analysis
- Customer analysis
- Predictive analytics reporting

6. Predictive Analytics

Python was used separately for:

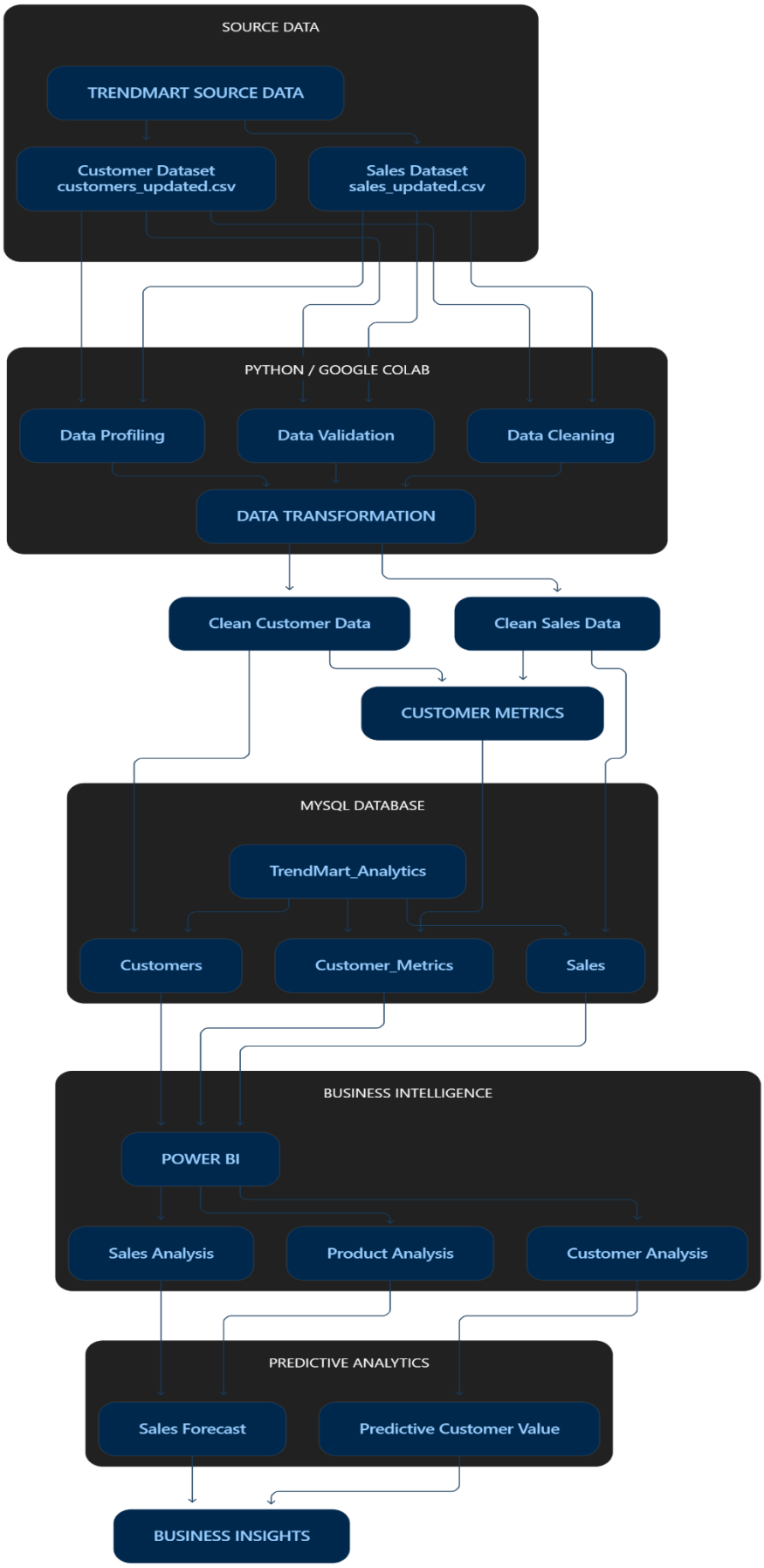
- Monthly sales forecasting
- Short-horizon predictive customer value analysis

7. Business Insights

The final solution provides insights into:

- Sales and revenue trends
- Product and category performance
- Customer purchasing behavior
- Customer segmentation
- Historical customer value
- Regional and payment behavior
- Forecasted sales
- Predicted future customer value

2.2 Architecture Flow



III. PROJECT APPROACH

3.1 Methodology

- **Source Data Identification**
  The available TrendMart customer and sales datasets were identified and reviewed to understand their structure, fields, and business meaning. The customer dataset contained **1,000 customer records and 6 columns**, while the sales dataset contained **2,705 transaction records and 9 columns.**

- **Data Extraction & Exploration**
  The CSV datasets were loaded into **Google Colab using Python and Pandas**. The datasets were profiled using:

  - Record counts
  - Column names
  - Data types
  - Missing-value analysis
  - Duplicate analysis
  - Unique-value analysis
  - Numerical statistics
  - Date-range analysis
  - Product analysis
  - Category analysis
  - Payment-method analysis
  - Customer coverage analysis

- **Data Validation**
  Validation checks were performed for:

  - Missing values
  - Duplicate records
  - Unique customer IDs
  - Unique transaction IDs
  - Customer-to-sales relationships
  - Numerical consistency
  - Date ranges
  - Quantity values
  - Unit prices
  - Transaction amounts

A numerical validation confirmed that:

Quantity × Unit Price = Total Amount

for the sales records.

- **Data Transformation**
  The cleaned datasets were transformed into analytical structures. Customer-level metrics were generated, including:

  - Transaction Frequency
  - Total Revenue
  - Average Transaction Value
  - Revenue Segment
  - Customer Segment

Customer segmentation was performed using customer purchasing frequency and revenue.

- **Data Staging & Loading**
  The cleaned datasets were exported as CSV files and loaded into the **TrendMart_Analytics** MySQL database. The main tables loaded were:

  - Customers
  - Sales
  - Customer_Metrics

- **MySQL Analysis**

SQL queries were used to analyze:

- Overall revenue
- Monthly revenue
- Product performance
- Category performance
- Payment-method performance
- Regional performance
- Customer segments
- Customer purchasing frequency
- Historical customer value
- Revenue contribution

- **Power BI Data Modeling & Reporting**
  The MySQL tables were connected to Power BI. Relationships were established between:

  - Customers and Sales
  - Customers and Customer_Metrics
  - DateTable and Sales

DAX measures were created and organized in a dedicated **TrendMart Measures** display folder. Four Power BI dashboard pages were developed.

- **Predictive Analytics**

Python was used to develop:

- A monthly sales forecasting model
- A short-horizon predictive customer value model

- **Business Analysis**
  The completed dashboards were used to analyze sales performance, products, categories, customer segments, customer value, and predictive trends.

3.2 Technology Stack

Component	Technology	Purpose
Source Data	CSV	Customer and sales transaction data
Data Processing	Python, Pandas	Data exploration, cleaning and transformation
Development Environment	Google Colab	Python-based data processing and modeling
Staging	CSV	Cleaned and transformed datasets
Data Warehouse / Database	MySQL	Structured analytical storage
Database Tool	MySQL Workbench	Database creation, loading and SQL analysis
Visualization	Power BI	Interactive dashboards and reporting
Calculations	DAX	KPI and analytical measures
Forecasting	Python, Scikit-learn	Sales forecasting
Predictive Analytics	Python, Scikit-learn	Short-horizon predictive customer value

IV. SAP SOURCE DATA & TABLE MAPPING

4.1 Source Data

The project uses two primary source datasets.

- **Customer Dataset : File:** customers_updated.csv

```
customers.head()
```

	Customer_ID	Name	Age	Gender	Location	Total_Spent
0	0583d6	Kevin Hanna	22	Female	Los Angeles	4179
1	4c0f1b	Marisa Sullivan	60	Other	Los Angeles	1683
2	00b49a	Shawn Gonzales	45	Female	Houston	358
3	e093b6	Charles Mejia	68	Male	New York	1541
4	7c817c	Kimberly Francis	43	Female	Los Angeles	2911

Records: 1,000

- **Sales Dataset: File:** sales_updated.csv

```
sales.head()
```

	Transaction_ID	Date	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Amount	Payment_Method
0	f3daa723	2024-01-01 00:00:00	0583d6	Clothing	Sofa	4	412	1648	Credit Card
1	9e5061b5	2024-01-01 01:00:00	4c0f1b	Home Decor	Novel	3	201	603	Cash
2	a5da9276	2024-01-01 02:00:00	00b49a	Home Decor	T-Shirt	1	480	480	Credit Card
3	b72e138c	2024-01-01 03:00:00	e093b6	Books	Laptop	2	16	32	Credit Card
4	c6010e46	2024-01-01 04:00:00	7c817c	Home Decor	Laptop	4	342	1368	Debit Card

Records: 2,705

4.2 Target Analytical Tables

```
CUSTOMERS DATASET
Shape: (1000, 6)
Null values: 0
Duplicate rows: 0
Unique Customer IDs: 1000

SALES DATASET
Shape: (2705, 9)
Null values: 0
Duplicate rows: 0
Unique Transaction IDs: 2705
Unique Customer IDs: 1000

CUSTOMER METRICS
Shape: (1000, 6)
Null values: 0
Unique Customer IDs: 1000
```

V. DATA EXPLORATION & PRIMARY KEY VALIDATION

5.1 Data Exploration

Initial data profiling was performed using Python and Pandas.

The analysis included:

- Record counts
- Column names
- Data types
- Missing-value analysis
- Duplicate analysis
- Unique customer analysis
- Unique transaction analysis
- Numerical field analysis
- Date-range analysis
- Product analysis
- Category analysis
- Payment-method analysis
- Location analysis

The profiling confirmed that the cleaned customer dataset contained **1,000 records**, while the sales dataset contained **2,705 records**.

5.2 CustomerID Validation

The customer dataset contained:

- **1,000 rows**
- **6 columns**
- **1,000 unique Customer IDs**
- **0 missing values**
- **0 duplicate rows**

Customer distribution:

- Female: 335
- Male: 330
- Other: 335

Locations:

- Chicago: 209
- New York: 207
- Houston: 199
- Los Angeles: 197
- Miami: 188

Age statistics:

- Mean age: approximately 43.57 years
- Minimum age: 18
- Maximum age: 69

5.3 Sales Data Validation

The sales dataset contained:

- **2,705 rows**
- **9 columns**
- **2,705 unique transactions**
- **1,000 unique customers**
- **0 missing values**
- **0 duplicate rows**

Sales date range:

1 January 2024 to 14 April 2024

The sales data contained five product categories and five main products.

5.4 Numerical Validation

The following numerical consistency rule was validated:

Quantity × Unit Price = Total Amount

The validation confirmed consistency between the quantity, unit price, and transaction amount fields.

Key statistics:

- Total Revenue: **₹1,782,148**
- Total Quantity Sold: **6,927**
- Total Transactions: **2,705**
- Average Transaction Value: **₹658.83**

5.5 Customer-Level Metrics

Customer-level metrics were generated from the sales transactions.

The metrics included:

- Transaction Frequency
- Total Revenue
- Average Transaction Value
- Revenue Segment
- Customer Segment

The customer segmentation resulted in:

Customer Segment	**Customers**
Occasional	469
High Value	287
High Value Frequent	213
Frequent	31
Total	**1,000**

The customer metrics dataset contained:

- 1,000 rows
- 6 analytical columns
- 1,000 unique Customer IDs
- 0 missing values

VI. MYSQL DATA WAREHOUSE IMPLEMENTATION & DATA LOADING

6.1 MySQL DataBase Implementation

After data cleaning and transformation, the analytical datasets were loaded into the **TrendMart_Analytics** MySQL database. MySQL was used to provide a structured relational environment for storing customer, transaction, and customer-level analytical data.

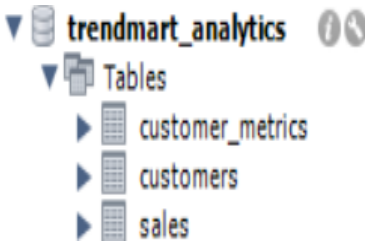


Figure 6.1: Tables created in MySQL Workbench

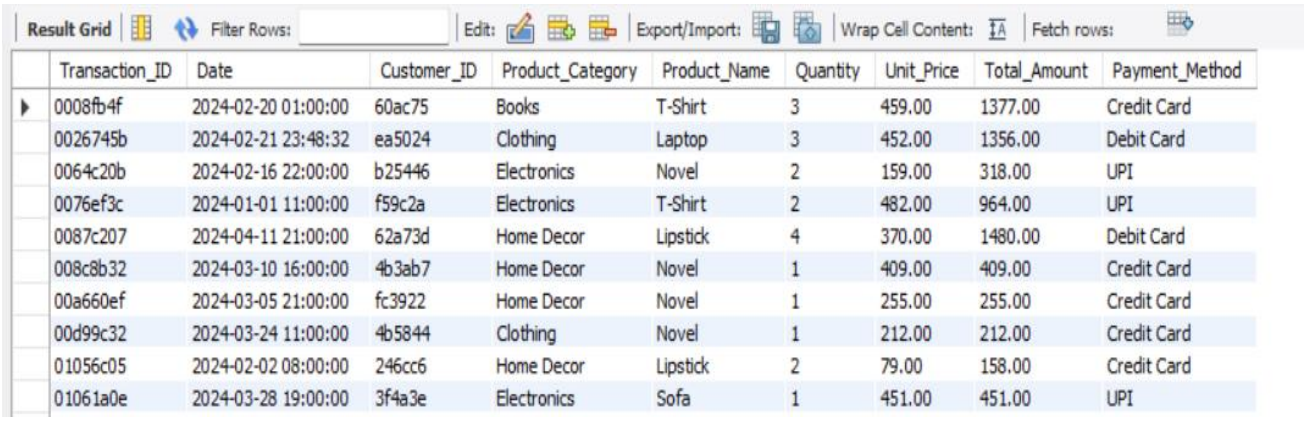
6.2 Data Warehouse Structure

- **Customers:** Stores customer master information such as:

Result Grid | Filter Rows: | Edit: | Export/Import:

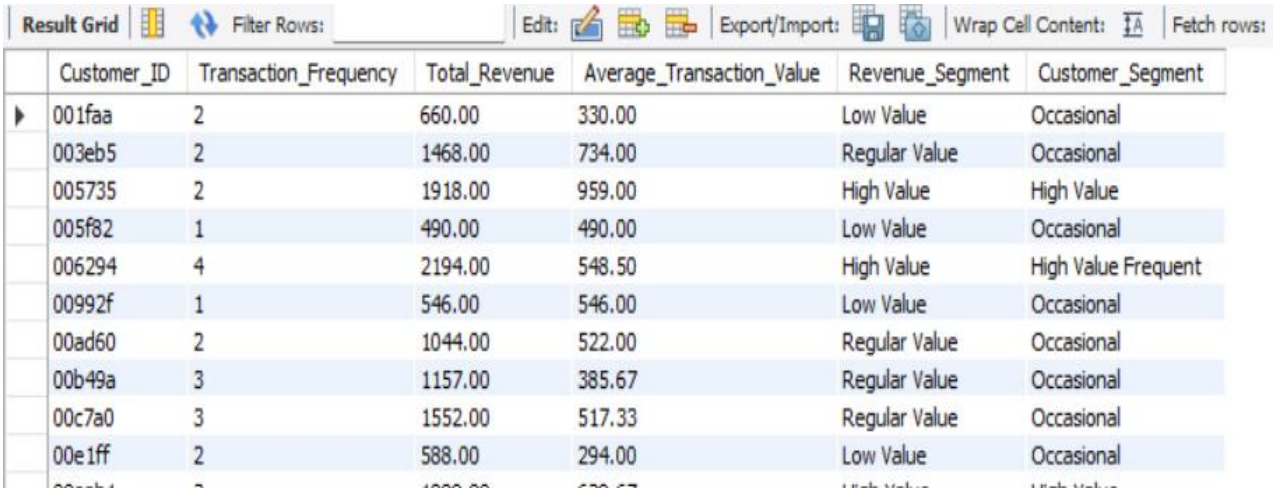
Customer_ID	Name	Age	Gender	Location	Total_Spent
001faa	Shirley Allen	32	Female	Houston	3500.00
003eb5	Bailey Thompson	52	Male	Miami	1965.00
005735	Aaron Barnes	35	Male	Chicago	310.00
005f82	Alexis Barnes	36	Female	Miami	4669.00
006294	Rebecca Clark	29	Female	Miami	102.00
00992f	Heather Torres	50	Other	Chicago	639.00
00ad60	Hunter Johnston	55	Female	Miami	4052.00
00b49a	Shawn Gonzales	45	Female	Houston	358.00
00c7a0	Steven Hernandez Jr.	42	Female	New York	689.00
00e1ff	Daniel Rhodes	30	Male	Houston	4794.00
00eab1	Samantha Jones	62	Other	Chicago	436.00

- **Sales :** Stores transaction-level information such as:



Transaction_ID	Date	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Amount	Payment_Method
0008fb4f	2024-02-20 01:00:00	60ac75	Books	T-Shirt	3	459.00	1377.00	Credit Card
0026745b	2024-02-21 23:48:32	ea5024	Clothing	Laptop	3	452.00	1356.00	Debit Card
0064c20b	2024-02-16 22:00:00	b25446	Electronics	Novel	2	159.00	318.00	UPI
0076ef3c	2024-01-01 11:00:00	f59c2a	Electronics	T-Shirt	2	482.00	964.00	UPI
0087c207	2024-04-11 21:00:00	62a73d	Home Decor	Lipstick	4	370.00	1480.00	Debit Card
008c8b32	2024-03-10 16:00:00	4b3ab7	Home Decor	Novel	1	409.00	409.00	Credit Card
00a660ef	2024-03-05 21:00:00	fc3922	Home Decor	Novel	1	255.00	255.00	Credit Card
00d99c32	2024-03-24 11:00:00	4b5844	Clothing	Novel	1	212.00	212.00	Credit Card
01056c05	2024-02-02 08:00:00	246cc6	Home Decor	Lipstick	2	79.00	158.00	Credit Card
01061a0e	2024-03-28 19:00:00	3f4a3e	Electronics	Sofa	1	451.00	451.00	UPI

- **Customer_Metrics:** Stores derived customer-level analytical metrics:



Customer_ID	Transaction_Frequency	Total_Revenue	Average_Transaction_Value	Revenue_Segment	Customer_Segment
001faa	2	660.00	330.00	Low Value	Occasional
003eb5	2	1468.00	734.00	Regular Value	Occasional
005735	2	1918.00	959.00	High Value	High Value
005f82	1	490.00	490.00	Low Value	Occasional
006294	4	2194.00	548.50	High Value	High Value Frequent
00992f	1	546.00	546.00	Low Value	Occasional
00ad60	2	1044.00	522.00	Regular Value	Occasional
00b49a	3	1157.00	385.67	Regular Value	Occasional
00c7a0	3	1552.00	517.33	Regular Value	Occasional
00e1ff	2	588.00	294.00	Low Value	Occasional

VII. POWER BI DATA MODEL & KPI DEVELOPMENT

7.1 Power BI Data Model

The MySQL database was connected to Power BI.

The following relationships were established:

- **Customers → Sales**
- **Customers → Customer_Metrics**
- **DateTable → Sales**

A dedicated **DateTable** was created for date-based sales analysis.

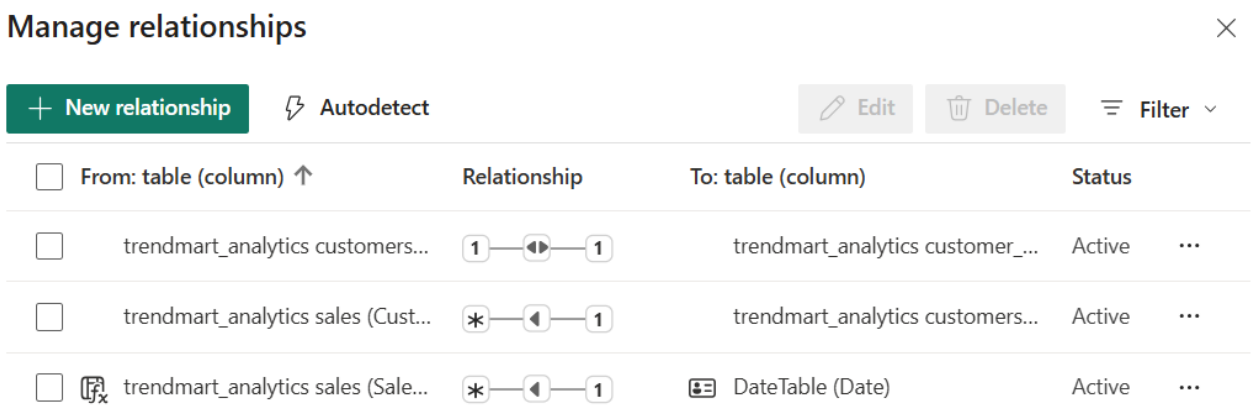


Figure 7.1: Power BI relational data model

7.2 DAX Measures & KPI Development

DAX measures were created in Power BI to calculate the key performance indicators required for the analysis. DAX measures were created in Power BI and organized into the dedicated **TrendMart Measures** display folder.

The completed measures include:

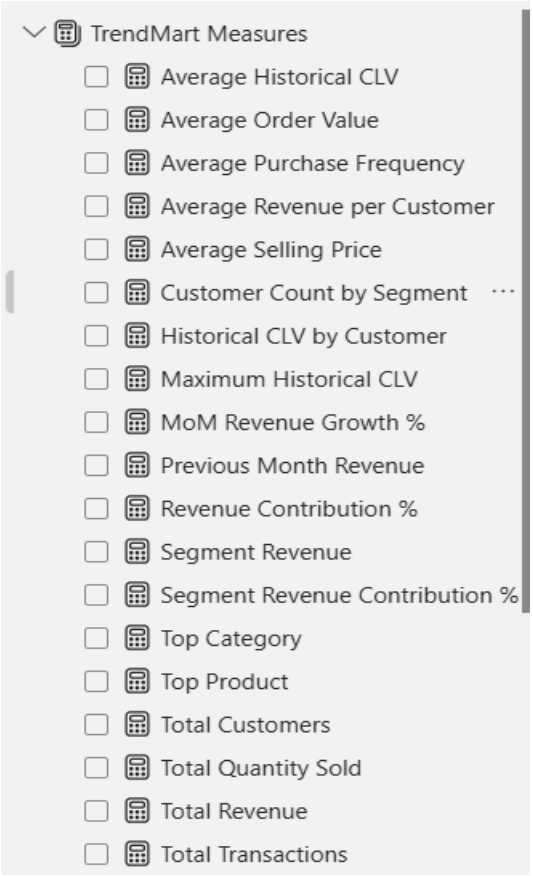


Figure 7.2: DAX measures organized in TrendMart Measures Folder

7.3 Power BI Data Model View

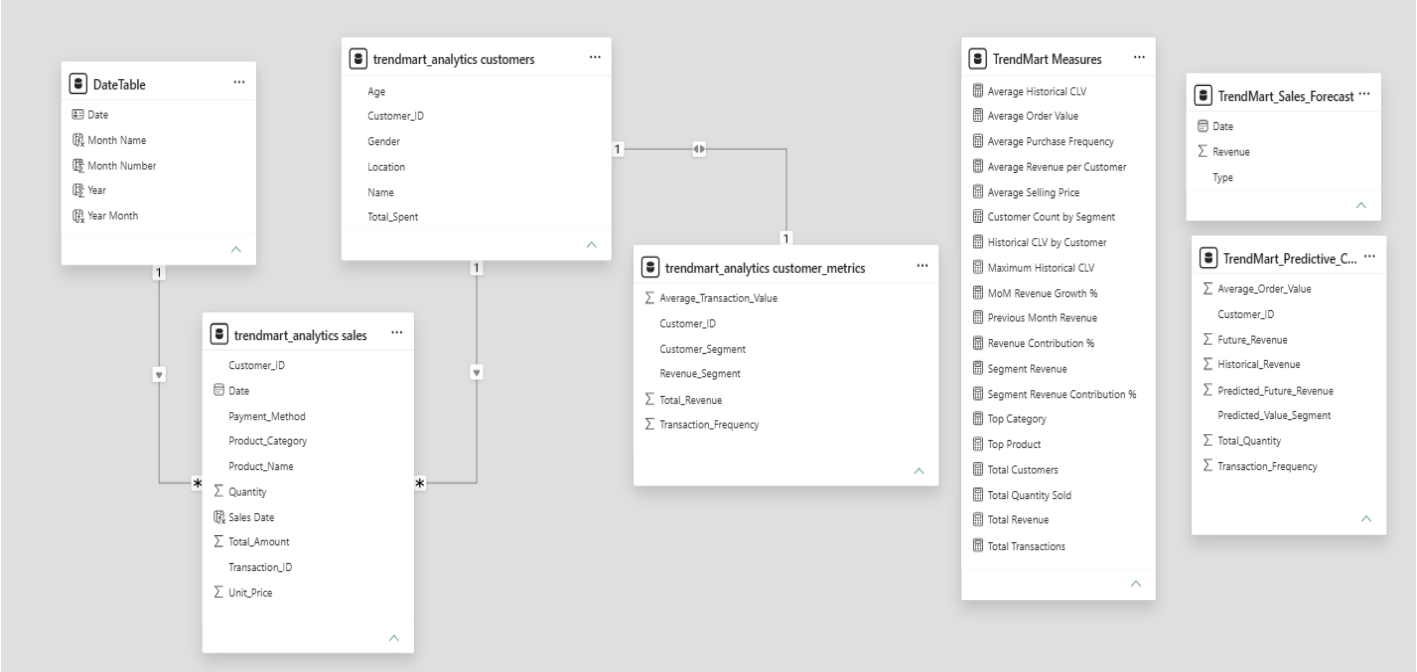


Figure 7.3: TrendMart Data Model View

VIII. POWER BI DASHBOARDS & ANALYTICAL INSIGHTS

8.1 Sales Overview

The **TrendMart – Sales Overview** page provides an overall view of sales performance.

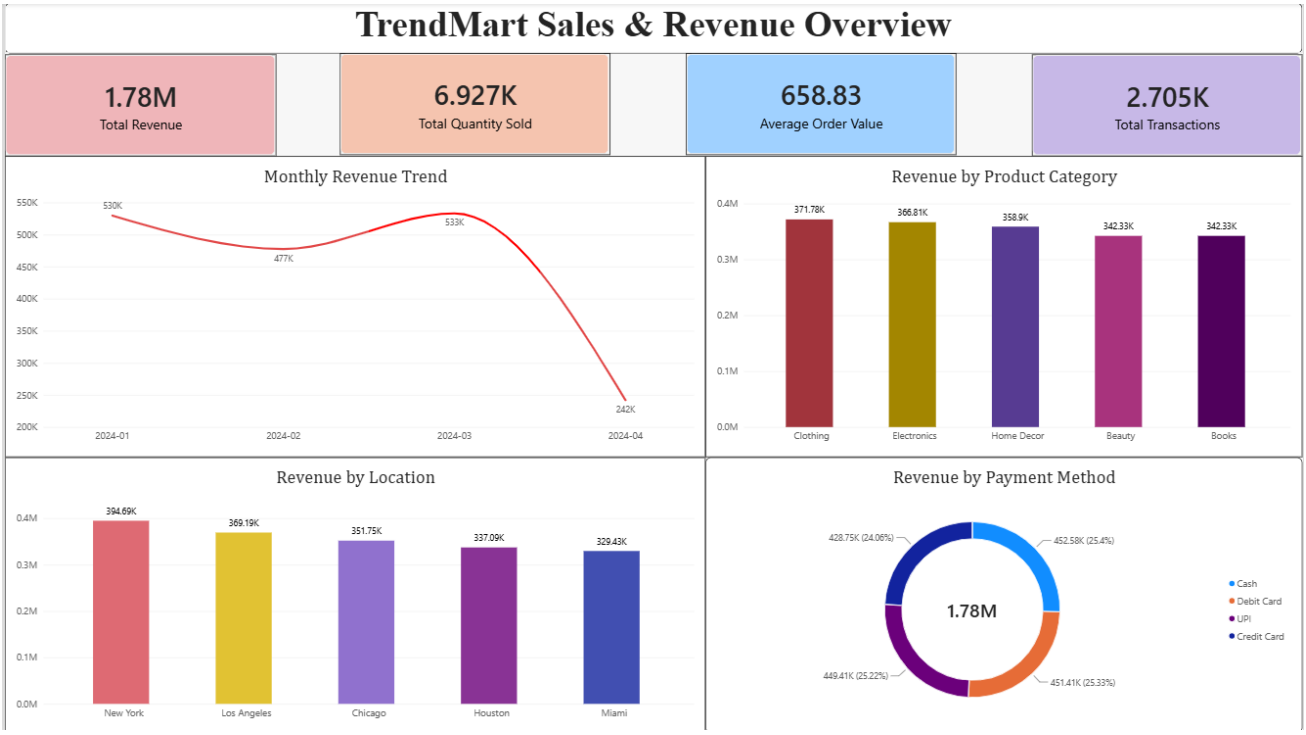


Figure 8.1: : TrendMart Sales Overview dashboard

8.2 Product & Category Analysis

The **TrendMart – Product & Category Analysis** page focuses on product and category performance.

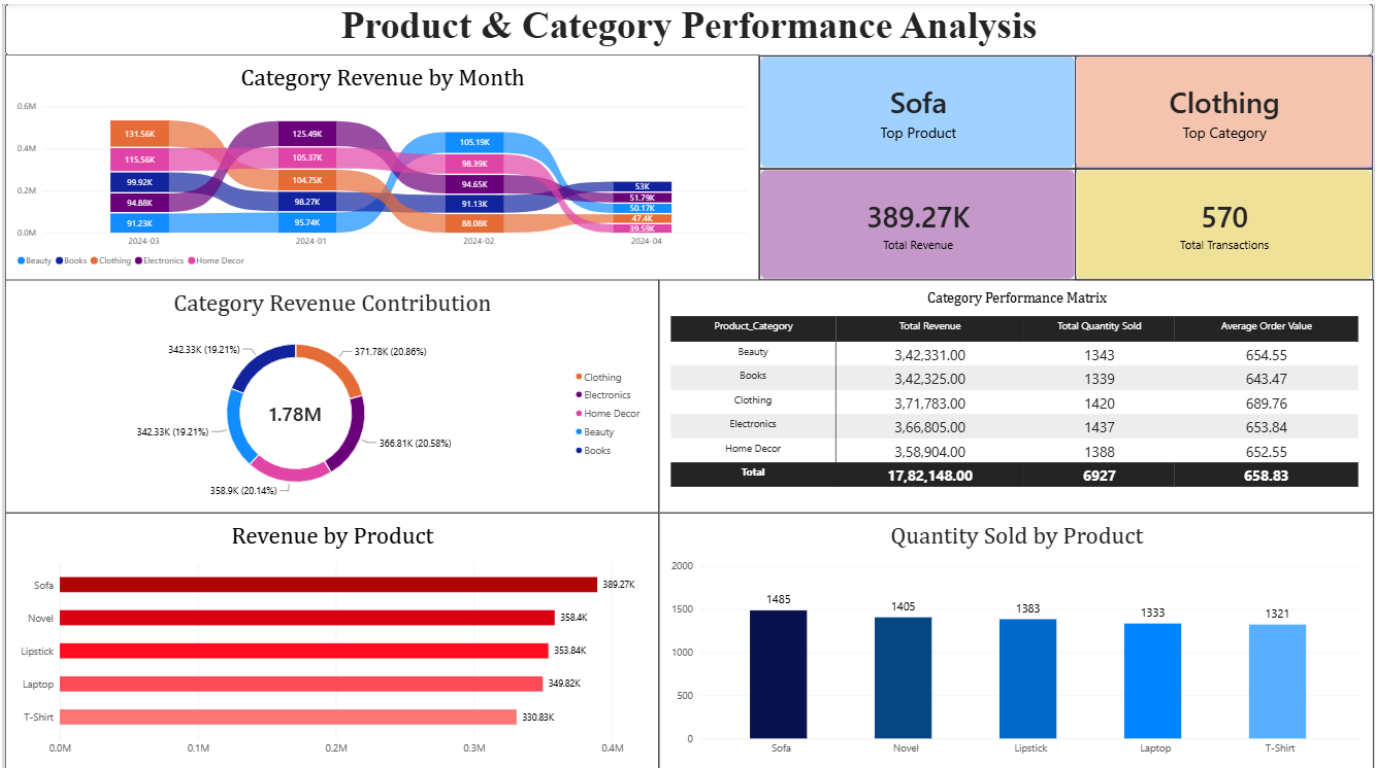


Figure 8.2: TrendMart Product & Category Analysis dashboard

8.3 Customer & CLV Analysis

The **TrendMart – Customer & CLV Analysis** page focuses on customer purchasing behavior and historical customer value.

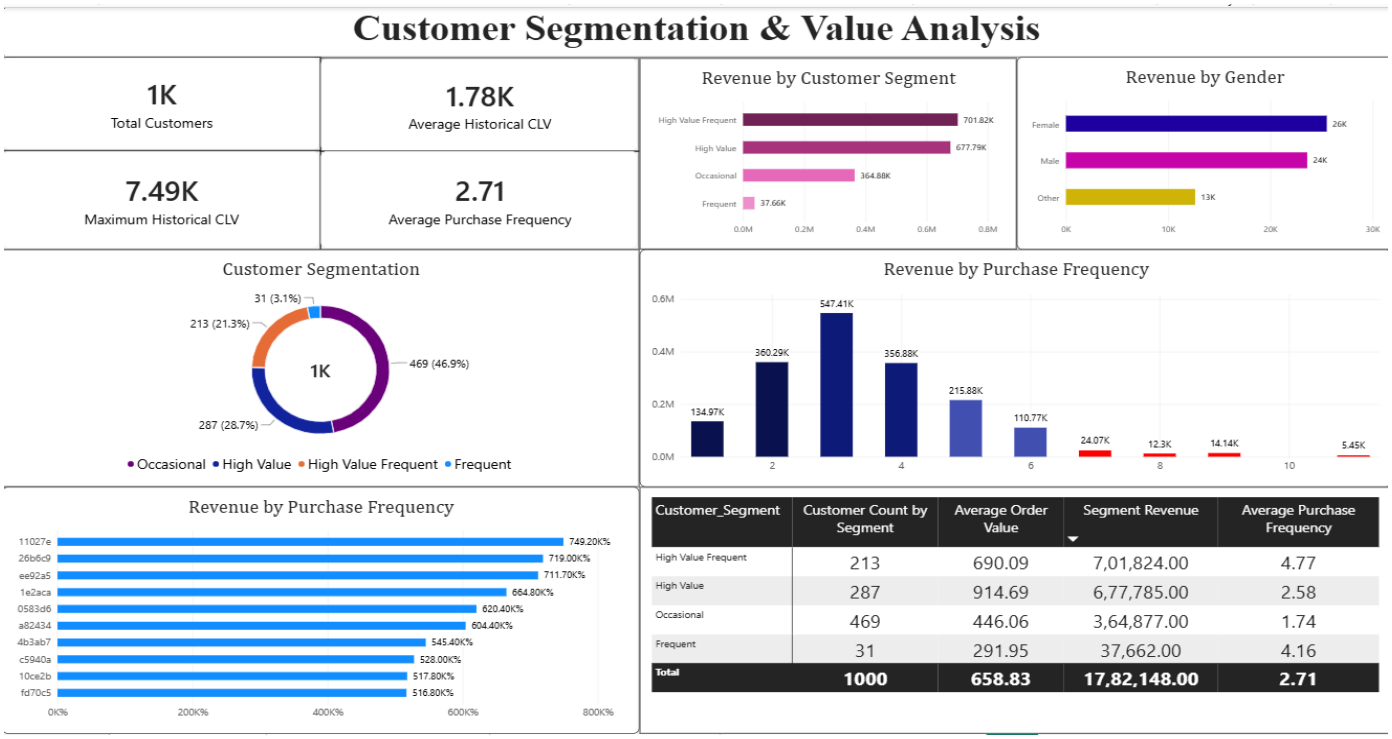


Figure 8.3 TrendMart Customer & CLV Analysis dashboard

8.4 Predictive Sales & Customer Insights

This page combines the Python forecasting and predictive customer-value results.

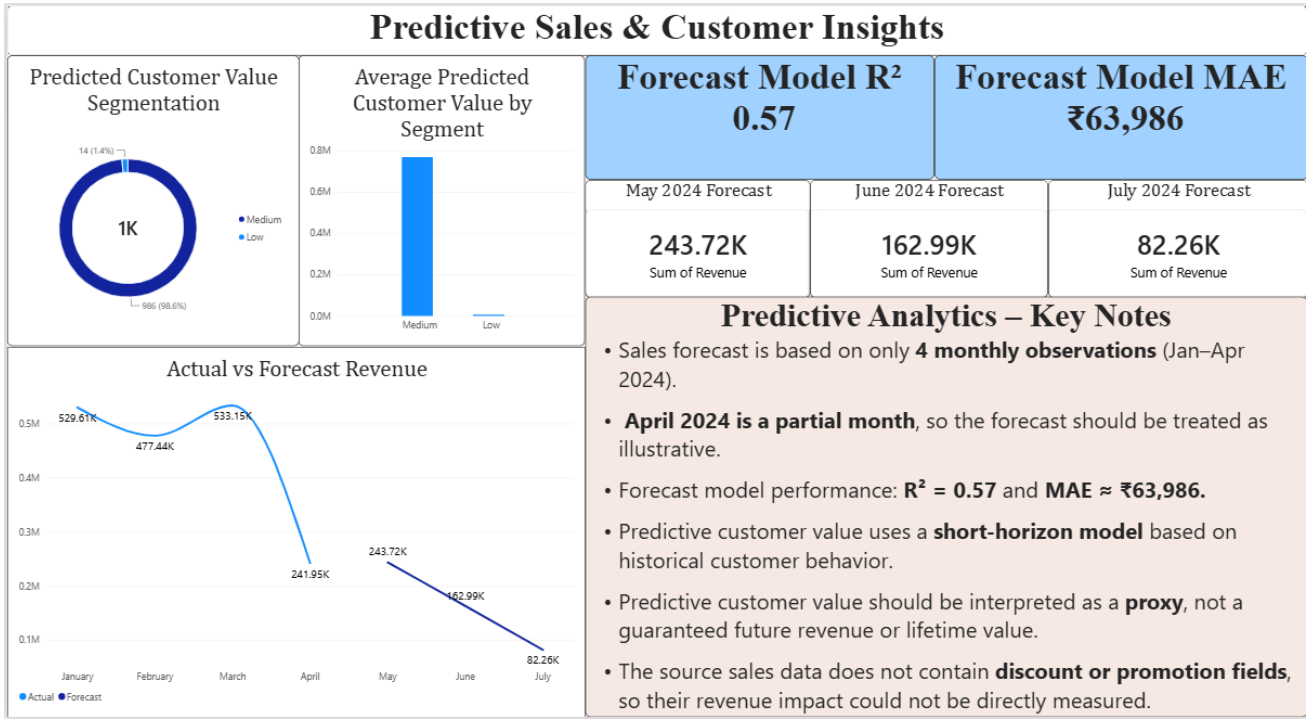


Figure 8.4: TrendMart Predictive Sales & Customer Insights dashboard

IX. BUSINESS ANALYTICAL INSIGHTS

9.1 Sales Insights

The TrendMart sales analysis provides an overview of revenue, transactions, quantity sold, and monthly sales trends.

Key observations:

- Total revenue: **₹1.78M**
- Total transactions: **2,705**
- Total quantity sold: **6,927**
- March recorded the highest revenue among the available complete months.
- April data is partial and covers transactions only through 14 April 2024.

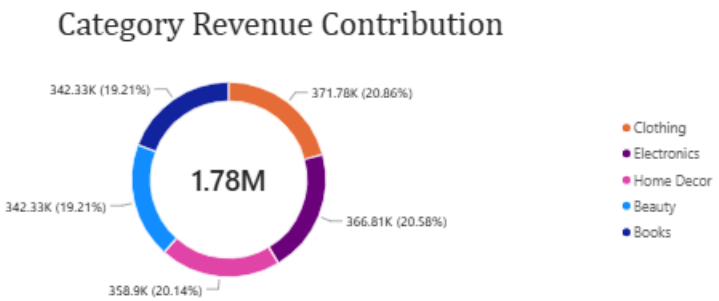


9.2 Product & Category Insights

Product and category analysis was performed to identify the products and categories contributing most to revenue.

Key observations:

- Sofa was the highest-revenue product.
- Clothing was the highest-revenue category.
- Revenue was distributed relatively evenly across the five categories.

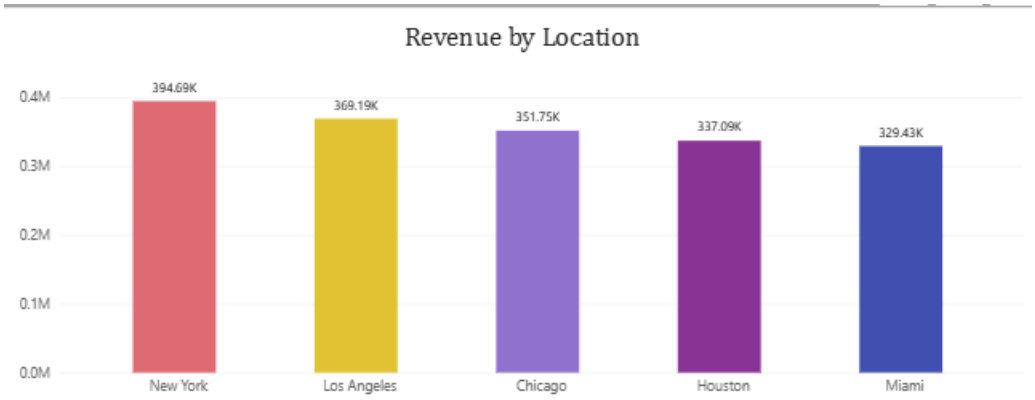


9.3 Regional Insights

Revenue was analyzed across the five available customer locations.

Key observation:

New York recorded the highest revenue at ₹394,690, followed by Los Angeles and Chicago.

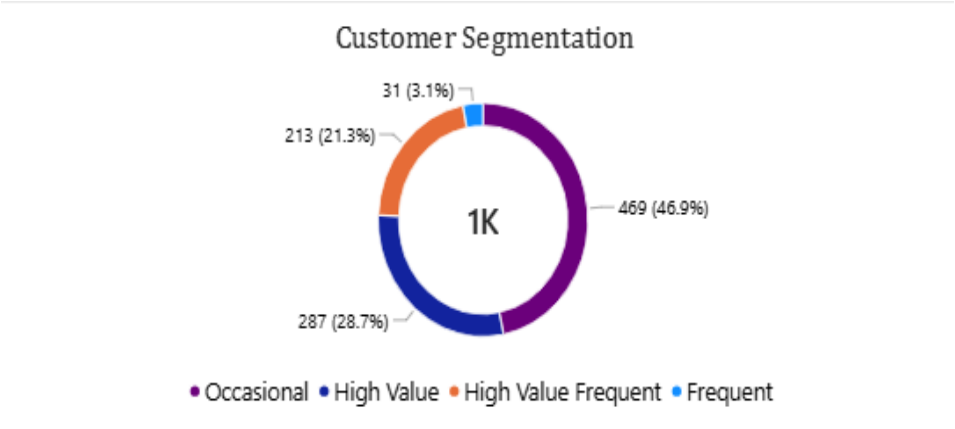


9.4 Customer Insights

Customer analysis was performed using transaction frequency, revenue, and customer segmentation.

Key observations:

- The dataset contains **1,000 customers**.
- Average purchase frequency was approximately **2.71 transactions per customer**.
- Customers were divided into **Occasional, Frequent, High Value, and High Value Frequent** segments.
- High Value and High Value Frequent customers contributed approximately **77.41% of total revenue**.

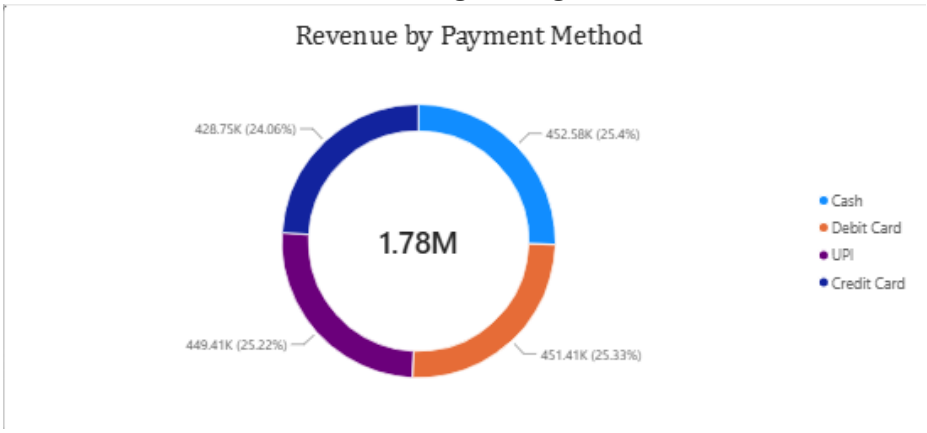


9.5 Payment Method Insights

Revenue was also analyzed according to payment method.

Key observation:

Revenue was relatively evenly distributed across Cash, Debit Card, UPI, and Credit Card transactions, with Cash recording the highest revenue.

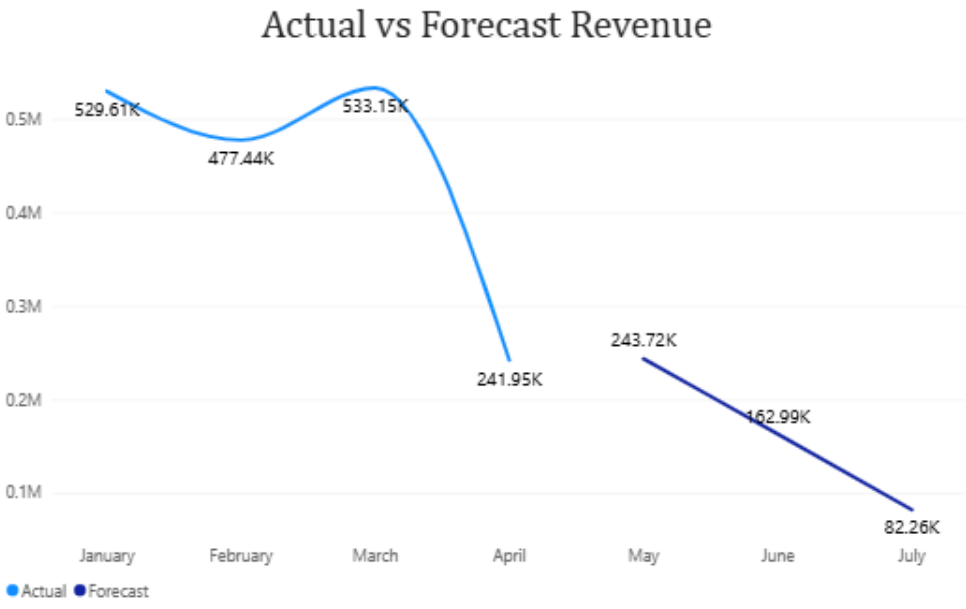


9.6 Predictive Insights

Python-based predictive analysis was incorporated into the Power BI dashboard.

Key observations:

- The sales forecasting model generated projections for **May–July 2024**.
- The forecast model achieved **$R^2 = 0.57$** with **MAE ≈ ₹63,986**.
- Predictive customer value was modeled using historical customer behavior.
- The predictive customer-value model achieved a test **$R^2 \approx 0.033$**, so the output is treated as a **short-horizon predictive CLV proxy**.



X. DATA LIMITATIONS

The following limitations were identified during the project and were explicitly documented rather than introducing unsupported assumptions.

1. Limited Sales History

The sales dataset covers only:

1 January 2024 to 14 April 2024

Therefore, the available time period is limited for long-term trend analysis.

2. Partial April Data

April contains transactions only through **14 April 2024**.

Therefore, April should not be directly compared with complete months without considering the incomplete observation period.

3. No Discount or Promotion Fields

The source sales dataset does not contain:

- Discount amount
- Discount percentage
- Promotion ID
- Campaign information
- Original price versus discounted price

Therefore, the direct revenue impact of discounts and promotions could **not** be quantitatively measured.

4. Historical Customer Value vs Predictive CLV

The historical customer value calculated in Power BI is based on transaction-derived revenue.

It represents **historical customer value**, not a guaranteed future lifetime value.

The predictive model was therefore documented separately as a **Short-Horizon Predictive CLV Proxy**.

5. Predictive Model Limitations

The predictive customer-value model achieved a relatively low test R^2 of approximately **0.033**.

Therefore, the predictions should be treated as analytical estimates rather than guaranteed future customer revenue.

6. Forecast Model Limitations

The sales forecast was developed using only four monthly observations.

Therefore, the forecast provides an **illustrative predictive trend** rather than a robust long-term forecasting model.

XI. CONCLUSION

11.1 Conclusion

The TrendMart case study implemented an end-to-end analytics solution using **Python, MySQL, and Power BI**.

The customer and sales datasets were extracted, explored, validated, cleaned, transformed, and loaded into a structured MySQL analytical database.

SQL analysis was performed to understand sales, product, category, customer, regional, payment, and customer-value patterns.

The MySQL data was then connected to Power BI, where DAX measures, relationships, and interactive dashboards were developed.

Python was additionally used to create a sales forecasting model and a short-horizon predictive customer value model.

The final solution provides a structured view of **sales performance, product performance, customer behavior, customer segmentation, historical customer value, and predictive insights**.

11.2 Key Outcomes

- Implemented an end-to-end ETL pipeline using **Python and MySQL**.
- Profiled and validated **1,000 customer records**.
- Profiled and validated **2,705 sales transactions**.
- Created structured MySQL tables for **Customers, Sales, and Customer_Metrics**.
- Performed SQL-based business analysis.
- Created a Power BI data model with customer, sales, and date relationships.
- Developed DAX-based sales and customer KPIs.
- Created **four Power BI dashboard pages**.
- Performed product and category performance analysis.
- Performed customer segmentation and historical customer value analysis.
- Developed a Python-based sales forecasting model.
- Developed a short-horizon predictive customer value model.
- Documented limitations related to limited historical data, partial April data, missing promotion fields, and predictive-model performance.
- Avoided unsupported assumptions regarding discount and promotion impact.

11.2 Github Link

https://github.com/surabhi-0405/Case-Study-3-Sales-Revenue-Analysis-for-a-Small-Business-